Laboratory 2: From a Simple CGE Model to a Spatial CGE Network Economy

Mathematical Modelling of Economic Systems

1 Purpose and structure of the laboratory

The purpose of this laboratory is to build a spatial computable general equilibrium (SCGE) model step by step. Rather than starting from a large nonlinear system, we begin with a very small one-region CGE model and add one economic mechanism at a time.

The laboratory has the following progression:

1. a one-region CGE model;
2. a two-node SCGE model without transport costs;
3. iceberg transport costs;
4. Armington substitution between goods from different origins;
5. intermediate inputs and input–output linkages;
6. comparative-static experiments with productivity and transport costs.

Optional extensions: richer network structures, welfare measures, infrastructure investment, public financing, capital, economies of scale, migration, and multiple equilibria.

The laboratory contains six compulsory parts, worth **1 point each**, giving a maximum of **6 compulsory points**. Extensions A–C are worth **1 point each**. Advanced Extensions D–H are worth **2-3 points each**. A maximum of 6 bonus points can be awarded, even if the total value of the completed extensions exceeds 6 points. Extensions are awarded points only when they are implemented correctly, the numerical solution is verified, and the economic mechanisms are clearly interpreted using the model results.

A maximum of **6 bonus points** can be awarded, even if the total value of the completed extensions exceeds 6 points. The maximum score for the laboratory is therefore **12 points**.

2 General computational instructions

You may use any suitable software or programming language, such as Python, MATLAB, Julia, R, Mathematica, or GAMS. You should be able to explain how the model is implemented and verify that the equilibrium conditions are satisfied. Python is recommended, but not required. Useful Python tools include `scipy.optimize.brentq` and `scipy.optimize.root` for solving equilibrium conditions, and `numpy.linalg.solve` for linear systems.

All prices and wages must be positive. For multidimensional nonlinear systems, a convenient strategy is therefore to solve for logarithms of prices and wages. For example, if the solver variable is $x = \ln p$, recover the price as

$$p = e^x.$$

You may use AI tools for programming assistance. **However, you are responsible for checking your code, understanding every equation used in the model, and writing your own economic interpretation.**

For every equilibrium that you report, include numerical checks showing that the relevant market-clearing residuals are close to zero.

A numerical solver reporting "success" does not by itself imply that an economic equilibrium has been found. Solver success only means that the numerical algorithm has satisfied its own convergence criterion. After solving the model, always evaluate the equilibrium residuals independently. A solution should only be accepted if the residuals of the equations used to solve the model, as well as the market-clearing and accounting conditions used for verification, are sufficiently close to zero.

A recurring idea in the laboratory is that an equilibrium model is not just a set of formulas evaluated once. Some variables are *endogenous*: they must take values that make several markets and accounting identities hold simultaneously. The numerical solver therefore does not replace the economic model. It is only a device for finding the values that satisfy the equilibrium conditions you have specified.

It is useful to keep the model in separate blocks in your code. A recommended structure is to have functions for prices and unit costs, household demand, trade or Armington shares, production, market residuals, and reporting. This makes it easier to see which economic mechanism is responsible for a result and much easier to diagnose errors.

Throughout the laboratory, distinguish carefully between an *equation used to solve the model* and an *identity used to verify the solution.* General equilibrium models often contain one redundant market-clearing equation because of Walras' law. A good implementation should nevertheless calculate the omitted residual after the solution has been found. A small residual is an important check that the equations and accounting have been implemented consistently.

3 Part I: A simple one-region CGE model (1 point)

We begin with the smallest useful CGE model. There is one region, two goods $k \in \{1, 2\}$, one representative household, and one primary factor, labour.

This first model is intentionally simple so that most of the equilibrium can be understood without a computer. Its purpose is to introduce the logic that will remain in all later parts: firms transform scarce factors into output, households receive factor income and demand goods, prices connect the two sides of the economy, and equilibrium requires that all markets are mutually consistent. When space and trade are introduced later, these same blocks remain; only the number of prices, flows, and feedback channels increases.

We use the following notation:

- i: origin node, where a good is produced;
- j: destination node, where a good is consumed or used as an input;
- p_i: factor-gate (producer) price of the good produced in node i;
- p_{ij}: delivered price in destination j of the good from origin i;
- Z_{ij}: household consumption in node j of the good from node i;
- X^d_{ij}: intermediate demand in node j for the good from node i;
- t_{ij}: iceberg transport-cost markup from i to j;
- w_j: wage in node j;

- $\bar{L}_j$: labour endowment in node j.

3.1 Production

Sector k produces with labour only:

$$Y_k = A_k L_k, \tag{1}$$

where $A_k > 0$ is labour productivity.

The unit labour requirement is

$$a_k^L = \frac{1}{A_k}. \tag{2}$$

Under perfect competition and constant returns to scale, price equals unit cost:

$$p_k = \frac{w}{A_k}. \tag{3}$$

Equation (3) is a zero-profit condition. With constant returns to scale, there is no fixed cost and the cost of one additional unit is also the average cost. Under perfect competition a firm cannot charge a price above unit cost in equilibrium, while a price below unit cost would make production unprofitable. Productivity therefore enters the model directly through prices: higher A_k means that less labour is needed per unit of output and therefore lowers the competitive price, holding the wage fixed.

3.2 Household

The representative household has Cobb–Douglas preferences:

$$U = Z_1^{\alpha_1} Z_2^{\alpha_2}, \tag{4}$$

where

$$\alpha_1 + \alpha_2 = 1, \qquad \alpha_k > 0. \tag{5}$$

Labour supply is fixed at $\bar{L}$. Household income is

$$I = w\bar{L}. \tag{6}$$

The budget constraint is

$$p_1 Z_1 + p_2 Z_2 = I. \tag{7}$$

Cobb–Douglas demand is

$$Z_k = \frac{\alpha_k I}{p_k}. \tag{8}$$

The Cobb–Douglas specification makes expenditure shares transparent. The household spends the fraction α_k of its income on good k, so $p_k Z_k = \alpha_k I$. This simple property will be useful when we later move from a single economy to bilateral spatial flows.

3.3 Market clearing

Market clearing is what turns the individual decisions of firms and households into a general equilibrium. A quantity demanded by the household must ultimately be produced, and the labour used by all sectors together cannot exceed the fixed labour endowment. These conditions link quantities that would otherwise be determined independently.

Goods markets require

$$Y_k = Z_k, \qquad k = 1, 2. \tag{9}$$

Labour-market clearing requires

$$L_1 + L_2 = \bar{L}, \tag{10}$$

or equivalently

$$\frac{Y_1}{A_1} + \frac{Y_2}{A_2} = \bar{L}. \tag{11}$$

3.4 Numeraire

The model is homogeneous in nominal prices: if all prices and the wage are multiplied by the same positive constant, real demands and allocations do not change. Think of this as changing currency. The model can therefore determine only relative prices. One nominal variable must be fixed externally to set the unit in which all other prices are measured. This fixed variable is the *numeraire*.

Only relative prices matter. Set

$$w = 1. \tag{12}$$

Then

$$p_k = \frac{1}{A_k}.$$

3.5 Benchmark

Use

$$\bar{L} = 100, \qquad A_1 = 2, \qquad A_2 = 1, \qquad \alpha_1 = \alpha_2 = 0.5.$$

Your solution should satisfy approximately

$$w = 1, \qquad p_1 = 0.5, \qquad p_2 = 1,$$

$$Z_1 = Y_1 = 100, \qquad Z_2 = Y_2 = 50,$$

and

$$L_1 = L_2 = 50.$$

Tasks

1. Solve the benchmark analytically.
2. Implement the same model numerically.
3. Calculate the residuals of both goods markets and the labour market.
4. Explain why not all market-clearing equations are independent.
5. Explain what would happen if all nominal prices and the wage were multiplied by the same positive constant.

4 Part II: A two-node SCGE model without transport costs (1 point)

We now introduce space. There are two nodes,

$$\mathcal{N} = \{1, 2\}.$$

The key modeling change is that production and consumption now have a location. A good is identified by its node of origin, and households in both nodes may buy from either origin. In this part, there is still no physical cost of moving goods between nodes. The objective is therefore to isolate the general-equilibrium consequences of having several regional labor markets and several regional producers before transport frictions are added.

Each node contains

- one representative household;
- one representative competitive firm;
- a fixed labour endowment.

Each node produces one differentiated good. Thus, node 1 produces good 1 and node 2 produces good 2. At this stage, transportation is costless.

4.1 Production

The firm in node i uses local labour:

$$Y_i = A_i L_i^d. \tag{13}$$

Under perfect competition,

$$p_i = \frac{w_i}{A_i}. \tag{14}$$

Labour is immobile between nodes and supply is fixed:

$$L_i^d = \bar{L}_i. \tag{15}$$

Hence output supply is

$$Y_i = A_i \bar{L}_i. \tag{16}$$

Because each node has only one firm and labour cannot move between nodes, the physical quantity supplied by a node is fixed by its productivity and labour endowment in this simple version. The relative wage does not determine how much labour the node supplies; instead, it adjusts the value of the node's output so that demand for that output is consistent with supply. This is why wages play a central equilibrating role in Part II.

4.2 Households

The household in destination node j consumes goods from both origins. Use Cobb–Douglas utility in logarithmic form:

$$U_j = \sum_{i \in \mathcal{N}} \beta_{ij} \ln Z_{ij}, \tag{17}$$

where

$$\beta_{ij} > 0, \qquad \sum_{i \in \mathcal{N}} \beta_{ij} = 1. \tag{18}$$

The first index in Z_{ij} denotes the *origin*; the second denotes the *destination*.

Household income in node j is

$$I_j = w_j \bar{L}_j. \tag{19}$$

With no transport costs, the household budget is

$$\sum_i p_i Z_{ij} = I_j. \tag{20}$$

Demand is

$$Z_{ij} = \frac{\beta_{ij} I_j}{p_i}. \tag{21}$$

Thus the household spends a fixed fraction β_{ij} of its income on the good from origin i:

$$p_i Z_{ij} = \beta_{ij} I_j. \tag{22}$$

The matrix β therefore has a direct spatial interpretation. A large β_{ij} means that households in destination j devote a large fraction of their income to the good produced in origin i. At this stage these expenditure shares are parameters rather than behavioural responses to relative prices. That assumption will be relaxed in Part IV.

4.3 Goods-market clearing

All output from node i must be consumed somewhere:

$$Y_i = \sum_{j \in \mathcal{N}} Z_{ij}. \tag{23}$$

Multiplying by p_i and using equations (14) and (16),

$$w_i \bar{L}_i = \sum_j \beta_{ij} w_j \bar{L}_j. \tag{24}$$

This value-market equation is useful for solving the model.

4.4 Numeraire and equilibrium

Set

$$w_1 = 1. \tag{25}$$

The remaining relative wage w_2 is determined by market clearing.

For example, define

$$F(w_2) = w_2 \bar{L}_2 - \sum_j \beta_{2j} w_j \bar{L}_j. \tag{26}$$

The equilibrium solves

$$F(w_2) = 0. \tag{27}$$

After solving for w_2, calculate prices, household demands, output, and all market-clearing residuals.

4.5 Benchmark

Use

$$\bar{L}_1 = \bar{L}_2 = 100, \qquad A_1 = A_2 = 1,$$

and

$$\begin{pmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{pmatrix} = \begin{pmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{pmatrix}.$$

The benchmark is symmetric. You should obtain

$$w_1 = w_2 = 1, \qquad p_1 = p_2 = 1,$$

and the consumption-flow matrix

$$Z = \begin{pmatrix} 60 & 40 \\ 40 & 60 \end{pmatrix},$$

where rows are origins and columns are destinations.

Tasks

1. Implement the two-node model.
2. Verify both goods markets.
3. Explain economically why wages, rather than quantities of labour, adjust across nodes.
4. Explain why one nominal variable must be normalized.
5. Show that one market-clearing condition is redundant once household budgets and all other equilibrium conditions hold.

5 Part III: Introduce iceberg transport costs (1 point)

We now introduce spatial trade costs.

This is the first part in which geography creates a real resource cost. A good bought in another node is more expensive not because a tariff transfers money to someone else, but because additional units of the good must be produced to make delivery possible. The iceberg formulation is convenient because no separate transport industry is needed: the transport service is represented by goods that are used up in transit.

Let

$$t_{ij} \geq 0 \tag{28}$$

denote the iceberg transport markup from origin i to destination j, with

$$t_{ii} = 0. \tag{29}$$

5.1 Delivered prices

The delivered price is

$$p_{ij} = p_i(1 + t_{ij}). \tag{30}$$

The buyer in node j pays p_{ij}, while p_i remains the factory-gate price.

Keep these two prices conceptually separate. The producer receives the factory-gate price p_i. The buyer faces the delivered price p_{ij}, which includes the resources needed for transportation. Confusing these prices is one of the most common accounting errors in spatial equilibrium models.

Household demand becomes

$$Z_{ij} = \frac{\beta_{ij} I_j}{p_{ij}} = \frac{\beta_{ij} I_j}{p_i (1 + t_{ij})}. \tag{31}$$

Composite price index and real consumption

Although Cobb–Douglas expenditure shares are fixed in this part, we can still construct a consumer price index and a measure of real household consumption. This will allow the welfare effects of transport-cost changes to be compared with the later Armington models.

Define the Cobb–Douglas consumption composite in node j as

$$Q_j = \prod_{i \in \mathcal{N}} \left(\frac{Z_{ij}}{\beta_{ij}} \right)^{\beta_{ij}}, \qquad \sum_i \beta_{ij} = 1. \tag{32}$$

The corresponding unit-expenditure price index is

$$P_j^{CD} = \prod_{i \in \mathcal{N}} p_{ij}^{\beta_{ij}}. \tag{33}$$

Since Cobb–Douglas demand implies

$$p_{ij} Z_{ij} = \beta_{ij} I_j, \tag{34}$$

substituting the demand functions into equation (32) gives

$$Q_j = \frac{I_j}{P_j^{CD}}. \tag{35}$$

Thus P_j^{CD} measures the cost of the household consumption composite and Q_j provides a simple real-consumption welfare measure.

A reduction in transport costs lowers the delivered prices p_{ij} and may therefore lower P_j^{CD}, increasing real consumption even when nominal household income is unchanged.

Notice, however, that the Cobb–Douglas expenditure shares remain fixed:

$$\frac{p_{ij} Z_{ij}}{I_j} = \beta_{ij}. \tag{36}$$

Transport costs can therefore affect quantities and real consumption in Part III, but they cannot induce households to shift nominal expenditure toward relatively cheaper origins. This limitation motivates the Armington specification introduced in Part IV.

5.2 Transport resource use

The quantity Z_{ij} is interpreted as the quantity that arrives in destination j.

To deliver Z_{ij}, the origin must produce

$$(1 + t_{ij}) Z_{ij}. \tag{37}$$

The amount of the good absorbed in transportation is therefore

$$T_{ij} = t_{ij} Z_{ij}. \tag{38}$$

Hence output from node i must satisfy

$$Y_i = \sum_j (Z_{ij} + T_{ij}) = \sum_j (1 + t_{ij}) Z_{ij}. \tag{39}$$

The monetary transport cost is

$$p_i T_{ij} = p_i t_{ij} Z_{ij}. \tag{40}$$

In this formulation, expenditure on transport services is therefore not a pure monetary transfer. It corresponds to real resource use in the economy: labour, fuel, vehicles, infrastructure use, and other inputs are required to move goods between regions. The iceberg formulation represents these resources in a simplified way by treating them as an additional amount of the transported good that is absorbed during transportation.

5.3 An important implication

Under Cobb–Douglas demand across origins,

$$p_{ij} Z_{ij} = \beta_{ij} I_j. \tag{41}$$

Therefore,

$$p_i (1 + t_{ij}) Z_{ij} = \beta_{ij} I_j. \tag{42}$$

The value-market equation remains

$$p_i Y_i = \sum_j \beta_{ij} I_j. \tag{43}$$

You should think carefully about this result. With fixed Cobb–Douglas expenditure shares, a higher transport cost reduces the delivered quantity but does not change the share of expenditure directed toward each origin.

This limitation motivates Part IV. The result is not a numerical accident. It follows from Cobb–Douglas preferences across origins: expenditure shares are fixed by assumption. A higher transport cost reduces the physical amount that arrives, but the household still devotes the same fraction of income to the origin. If the aim is to analyse how buyers switch suppliers when relative delivered prices change, the demand system itself must be changed. Armington CES demand provides exactly that margin of substitution.

5.4 Benchmark transport costs

Use

$$t_{11} = t_{22} = 0, \qquad t_{12} = t_{21} = 0.20.$$

With the symmetric benchmark from Part II, the equilibrium remains

$$w_1 = w_2 = 1, \qquad p_1 = p_2 = 1.$$

For each household:

- expenditure on the local good remains 60;
- expenditure on the imported good remains 40;
- imported quantity falls from 40 to $40/1.2 \approx 33.333$.

The transport input on each cross-node flow is therefore approximately

$$T_{12} = T_{21} \approx 6.667.$$

Tasks

1. Add t_{ij}, p_{ij}, and T_{ij} to your model.
2. Verify equation (39).
3. Compare $t = 0$ with $t = 0.05, 0.10, 0.20, 0.40$ and plot imported quantities against t.
4. Explain why nominal expenditure shares do not respond to transport costs.
5. Explain why this feature may be unrealistic when goods from different origins are close substitutes.

6 Part IV: Armington substitution (1 point)

We now replace the Cobb–Douglas allocation across origins by a CES Armington aggregator. Goods are differentiated by place of origin, but households can substitute between them when relative delivered prices change.

The Armington assumption is a standard way of allowing both regions to supply varieties that are similar but not identical. The elasticity η controls how strongly demand moves between origins. When η is large, small delivered-price differences can produce large changes in sourcing. When η is close to one, buyers are less willing to substitute between origins.

For household j, define the composite consumption good

$$Q_j = \left[\sum_{i \in \mathcal{N}} \beta_{ij}^{1/\eta} Z_{ij}^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)}, \tag{44}$$

where

$$\eta > 1 \tag{45}$$

is the elasticity of substitution between origins.

The weights satisfy

$$\beta_{ij} > 0, \qquad \sum_i \beta_{ij} = 1. \tag{46}$$

6.1 Armington price index

The corresponding unit-expenditure function is

$$P_j = \left[\sum_i \beta_{ij} p_{ij}^{1-\eta} \right]^{1/(1-\eta)}. \tag{47}$$

The price index P_j is the minimum expenditure needed to obtain one unit of the composite good in node j. It summarizes all origin-specific delivered prices into one cost-of-living measure. A transport-cost reduction can therefore improve welfare even if nominal household income does not change, because it lowers the price of the consumption composite.

Since the household spends all income on the composite good,

$$Q_j = \frac{I_j}{P_j}. \tag{48}$$

Thus Q_j can also be interpreted as a simple real-income welfare measure.

6.2 Origin expenditure shares

The expenditure share on the good from origin i is

$$s_{ij} = \frac{\beta_{ij} p_{ij}^{1-\eta}}{\sum_h \beta_{hj} p_{hj}^{1-\eta}}. \tag{49}$$

Equation (49) is the behavioural link between transport costs and trade patterns. If the delivered price from origin i rises relative to competing origins, its expenditure share falls. The strength of this response depends on η. This is why transport policy can now change not only quantities shipped but also where households choose to source their consumption.

The shares satisfy

$$\sum_i s_{ij} = 1. \tag{50}$$

Bilateral household demand is

$$Z_{ij} = \frac{s_{ij} I_j}{p_{ij}}. \tag{51}$$

6.3 Goods-market clearing

As before, iceberg resource use must be included:

$$Y_i = \sum_j (1 + t_{ij}) Z_{ij}. \tag{52}$$

Multiplying by p_i,

$$p_i Y_i = \sum_j s_{ij} I_j. \tag{53}$$

Using $p_i = w_i / A_i$ and $Y_i = A_i \bar{L}_i$,

$$w_i \bar{L}_i = \sum_j s_{ij}(w, p, t)\, w_j \bar{L}_j. \tag{54}$$

Unlike in Part III, the origin shares s_{ij} now depend on relative delivered prices. Transport costs can therefore redirect expenditure across origins.

6.4 Solution strategy

Set

$$w_1 = 1.$$

For a trial value of w_2:

1. compute $p_i = w_i / A_i$;
2. compute $p_{ij} = p_i (1 + t_{ij})$;
3. compute P_j;
4. compute s_{ij};
5. compute $I_j = w_j \bar{L}_j$;

6. compute $Z_{ij} = s_{ij} I_j / p_{ij}$;
7. evaluate the node-2 value-market residual

$$F(w_2) = w_2 \bar{L}_2 - \sum_j s_{2j} I_j.$$

Solve

$$F(w_2) = 0.$$

Then verify the omitted node-1 market equation.

6.5 Benchmark

Continue to use

$$\bar{L}_1 = \bar{L}_2 = 100, \qquad A_1 = A_2 = 1,$$
$$t_{12} = t_{21} = 0.20, \qquad t_{11} = t_{22} = 0,$$
$$\eta = 4,$$

and

$$\beta = \begin{pmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{pmatrix}.$$

By symmetry,

$$w_1 = w_2 = 1, \qquad p_1 = p_2 = 1.$$

The composite price index is approximately

$$P_1 = P_2 \approx 1.06345.$$

The local expenditure share is approximately

$$s_{11} = s_{22} \approx 0.72160,$$

and the import expenditure share is

$$s_{21} = s_{12} \approx 0.27840.$$

Delivered import quantity is approximately

$$Z_{\text{import}} = \frac{0.27840 \times 100}{1.2} \approx 23.20.$$

Real consumption is approximately

$$Q_1 = Q_2 = \frac{100}{1.06345} \approx 94.03.$$

Experiments

Perform the following experiments.

Experiment 1: Transport costs. Use

$$t \in \{0, 0.05, 0.10, 0.20, 0.40\}.$$

Plot

- import expenditure shares;
- import quantities;
- composite price indices;
- real consumption Q_j.

Experiment 2: Substitution elasticity. Repeat the transport-cost experiment for

$$\eta \in \{1.5, 2, 4, 8\}.$$

Explain why the trade response is stronger when η is large.

Experiment 3: Productivity asymmetry. Set

$$A_1 \in \{0.8, 1.0, 1.2, 1.5\}, \qquad A_2 = 1.$$

Investigate the effects on

$$w_2/w_1, \quad p_1/p_2, \quad s_{ij}, \quad Q_1, \quad Q_2.$$

Before running the model, state your expected effects.

7 Part V: Intermediate inputs and input–output linkages (1 point)

We now add intermediate inputs. This creates production-network feedbacks: a change in the price of the good from one node affects production costs in other nodes, which in turn changes their prices, demand, and output.

Previously, we assumed labor was the only input used in production, but we now expand the model to include intermediate goods purchased both from other firms and produced internally.

This step is important because many effects of transport infrastructure operate through firms rather than only through household consumption. If firms import inputs, a reduction in transport costs lowers their input-price index. This changes unit costs and producer prices, which then feeds back into household demand and into the input costs of other firms. The model therefore becomes a small production network rather than only a model of final trade.

To make this additional feedback manageable, we use a Cobb–Douglas production function in combination with an Armington composite, instead of the CES technology used in the reference article.

The main advantage is that expenditure on labour and intermediate inputs can be written as fixed cost shares of output value. This allows the production-value block to be solved as a linear input–output system for any trial vector of wages and prices.

7.1 Production technology

The firm in node j uses local labour and an Armington composite intermediate input:

$$Y_j = \phi_j \left(\frac{L_j^d}{\delta_{lj}} \right)^{\delta_{lj}} \left(\frac{X_j^d}{\delta_{xj}} \right)^{\delta_{xj}}, \tag{55}$$

where

$$\delta_{lj} + \delta_{xj} = 1, \qquad \delta_{lj} > 0, \qquad \delta_{xj} > 0. \tag{56}$$

The normalization in equation (55) makes the unit-cost function particularly simple.

The parameters δ_{lj} and δ_{xj} can be read as cost shares in competitive equilibrium. Increasing δ_{xj} makes the firm more dependent on purchased intermediate goods and therefore increases the importance of trade costs and production-network feedbacks.

7.2 Intermediate Armington composite

The intermediate composite used by the firm in node j is

$$X_j^d = \left[\sum_i \gamma_{ij}^{1/\eta_X} (X_{ij}^d)^{(\eta_X - 1)/\eta_X} \right]^{\eta_X/(\eta_X - 1)}, \tag{57}$$

where

$$\gamma_{ij} > 0, \qquad \sum_i \gamma_{ij} = 1, \qquad \eta_X > 1.$$

The corresponding intermediate-input price index is

$$P_j^X = \left[\sum_i \gamma_{ij} p_{ij}^{1-\eta_X} \right]^{1/(1-\eta_X)}. \tag{58}$$

The expenditure share on inputs from origin i is

$$s_{ij}^X = \frac{\gamma_{ij} p_{ij}^{1-\eta_X}}{\sum_h \gamma_{hj} p_{hj}^{1-\eta_X}}. \tag{59}$$

7.3 Unit cost and zero profit

The unit-cost function corresponding to (55) is

$$c_j = \frac{1}{\phi_j} w_j^{\delta_{lj}} (P_j^X)^{\delta_{xj}}. \tag{60}$$

Under perfect competition and constant returns to scale,

$$p_j = c_j. \tag{61}$$

7.4 Cost shares

Define the value of output in node j as

$$V_j = p_j Y_j. \tag{62}$$

Cobb–Douglas technology implies

$$w_j L_j^d = \delta_{lj} V_j, \tag{63}$$

and

$$P_j^X X_j^d = \delta_{xj} V_j. \tag{64}$$

Thus

$$L_j^d = \frac{\delta_{lj} V_j}{w_j}. \tag{65}$$

Labour-market clearing requires

$$\frac{\delta_{lj} V_j}{w_j} = \bar{L}_j. \tag{66}$$

7.5 Household demand

Household income remains

$$I_j = w_j \bar{L}_j. \tag{67}$$

Households use the Armington system from Part IV. To distinguish household and firm sourcing, denote household shares by

$$s_{ij}^H = \frac{\beta_{ij} p_{ij}^{1-\eta_H}}{\sum_h \beta_{hj} p_{hj}^{1-\eta_H}}. \tag{68}$$

The household price index is

$$P_j^H = \left[\sum_i \beta_{ij} p_{ij}^{1-\eta_H} \right]^{1/(1-\eta_H)}. \tag{69}$$

Household bilateral demand is

$$Z_{ij} = \frac{s_{ij}^H I_j}{p_{ij}}. \tag{70}$$

Household real consumption is

$$Q_j = \frac{I_j}{P_j^H}. \tag{71}$$

7.6 Intermediate bilateral demand

Total expenditure by firm j on the intermediate composite is

$$E_j^X = \delta_{xj} V_j. \tag{72}$$

Expenditure on the intermediate variety from origin i is

$$E_{ij}^X = s_{ij}^X \delta_{xj} V_j. \tag{73}$$

Thus the quantity arriving at firm j is

$$X_{ij}^d = \frac{s_{ij}^X \delta_{xj} V_j}{p_{ij}}. \tag{74}$$

7.7 Goods-market clearing

The factory in node i must produce enough output for both final and intermediate use, including iceberg losses:

$$Y_i = \sum_j (1 + t_{ij}) \left(Z_{ij} + X_{ij}^d \right). \tag{75}$$

Total transport-resource use on link ij is

$$T_{ij} = t_{ij} \left(Z_{ij} + X_{ij}^d \right). \tag{76}$$

Multiplying equation (75) by p_i gives a very useful value equation:

$$V_i = \sum_j \left[s_{ij}^H I_j + s_{ij}^X \delta_{xj} V_j \right]. \tag{77}$$

The first term is final demand. The second is intermediate demand.

7.8 The input–output block

For given wages and prices, the household and intermediate sourcing shares are known. The goods-market equations can therefore be written as a linear system in the production values V_i.

Define the final-demand vector

$$b_i = \sum_j s_{ij}^H I_j, \tag{78}$$

and the input–output matrix

$$M_{ij} = s_{ij}^X \delta_{xj}. \tag{79}$$

The goods-market equations can then be written as

$$V = b + MV, \tag{80}$$

or equivalently

$$(I - M)V = b. \tag{81}$$

Hence,

$$\boxed{V = (I - M)^{-1} b.} \tag{82}$$

The vector b represents production required to satisfy final household demand, while MV represents additional production required to supply intermediate inputs. The inverse $(I - M)^{-1}$ therefore captures both the direct effect of final demand and the indirect production generated through intermediate-input linkages.

An important feature of the present model is that M is not fixed. The sourcing shares s_{ij}^X depend on delivered prices, so changes in wages, producer prices, or transport costs also change the input–output structure of the economy.

7.9 Recommended equilibrium algorithm

The model now contains a useful nested computational structure. The outer nonlinear solver searches for prices and the relative wage. Conditional on any trial values, the inner input–output problem is linear and can be solved exactly. Keeping these two levels separate makes both the economics and the code easier to understand.

Set

$$w_1 = 1.$$

Use the nonlinear unknown vector

$$x = (p_1, p_2, w_2). \tag{83}$$

For every trial x:

1. Compute delivered prices:
$$p_{ij} = p_i(1 + t_{ij}).$$
2. Compute household price indices P_j^H and household shares s_{ij}^H.
3. Compute intermediate price indices P_j^X and intermediate shares s_{ij}^X.
4. Compute household incomes:
$$I_j = w_j \bar{L}_j.$$
5. Construct b and M, then solve
$$(I - M)V = b.$$
6. Compute unit costs:
$$c_j = \frac{1}{\phi_j} w_j^{\delta_{lj}} (P_j^X)^{\delta_{xj}}.$$
7. Compute labour demands:
$$L_j^d = \frac{\delta_{lj} V_j}{w_j}.$$
8. Return the three residuals

$$R_1 = p_1 - c_1, \tag{84}$$
$$R_2 = p_2 - c_2, \tag{85}$$
$$R_3 = L_2^d - \bar{L}_2. \tag{86}$$

Solve

$$R(x) = 0. \tag{87}$$

Do not rely only on the solver's convergence message. Evaluate the residual vector $R(x^*)$ at the reported solution and verify that

$$L_1^d - \bar{L}_1 \approx 0. \tag{88}$$

Also verify both physical goods-market equations (75).

7.10 Benchmark

Use

$$\bar{L}_1 = \bar{L}_2 = 100, \qquad \phi_1 = \phi_2 = 1,$$
$$\delta_{l1} = \delta_{l2} = 0.6, \qquad \delta_{x1} = \delta_{x2} = 0.4,$$
$$t_{12} = t_{21} = 0.20, \qquad t_{11} = t_{22} = 0,$$

and identical household and intermediate Armington parameters:

$$\eta_H = \eta_X = 4,$$

$$\beta = \gamma = \begin{pmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{pmatrix}.$$

The symmetric equilibrium should satisfy approximately

$$w_1 = w_2 = 1,$$

$$p_1 = p_2 \approx 1.04186,$$

$$V_1 = V_2 \approx 166.667,$$

and

$$L_1^d = L_2^d = 100.$$

The household and intermediate composite price indices are approximately

$$P_1^H = P_2^H = P_1^X = P_2^X \approx 1.10797.$$

Tasks

1. Implement the intermediate-input model.
2. Calculate the matrix M, the vector b, and the solution V.
3. Verify equation (82) numerically.
4. Verify zero profit, labour-market clearing, and physical goods-market clearing.
5. Explain the economic meaning of the matrix M.
6. Explain why a price shock in node 1 can now affect production costs in node 2 even if node 2's own technology is unchanged.

8 Part VI: Compare the models (1 point)

The purpose of the final compulsory part is to compare mechanisms, not merely numbers.

Because Parts III–V are nested versions of the same spatial economy, comparing the same transport-cost shock across them is more informative than running unrelated experiments. The aim is to identify which conclusions are robust and which appear only after a specific behavioural or production mechanism is introduced. Your discussion should therefore focus on causal chains, not only on whether a variable rises or falls.

Consider a reduction in symmetric inter-node transport costs from

$$t = 0.20$$

to

$$t = 0.10.$$

Analyse the same transport-cost reduction under:

1. the fixed-expenditure-share model from Part III;
2. the Armington model from Part IV;
3. the intermediate-input model from Part V.

The main purpose of this comparison is to understand how the model changes as new economic mechanisms are introduced.

For each model:

- reproduce the main benchmark results from the previous part;
- apply the transport-cost reduction;
- verify that the new equilibrium satisfies the relevant market-clearing and accounting conditions;
- report only the main variables needed to explain the result.

You do not need to report every endogenous variable. Focus on the variables that are most important for understanding how the model works, for example trade flows, relative wages, consumer prices, real consumption, and production.

Discuss the following:

1. Why is the response of trade different in Parts III and IV?
2. What new economic mechanism is introduced by the Armington specification?
3. What new mechanism is introduced by intermediate inputs in Part V?
4. Why can intermediate inputs strengthen the effect of a transport-cost reduction?
5. Which effects are direct consequences of lower transport costs, and which arise through general-equilibrium adjustment?
6. How does the interpretation of the same transport-cost shock change as the model becomes more detailed?

9 Submission requirements for the compulsory part

Submit a short report showing that you have worked through the compulsory models and understand how the different parts fit together.

The report should focus on the economic structure of the models rather than on producing large amounts of numerical output.

For Parts I–V:

- reproduce the requested benchmark results;
- show the main numerical checks needed to verify the equilibrium;
- answer the questions given in each part;
- briefly explain the economic mechanism introduced in that part.

For Part VI, compare the three spatial models and explain how the response to the same transport-cost reduction changes when additional mechanisms are introduced.

Use a small number of clear tables and figures where they help the discussion. Avoid printing complete solver output or large matrices unless they are directly needed for the analysis.

Your discussion should make clear that you understand:

- which variables are exogenous and endogenous;
- which equations determine the equilibrium;
- why a numeraire is required;
- how market clearing links households and firms;
- how transport costs enter delivered prices and resource use;
- how Armington substitution changes trade responses;
- how intermediate inputs create additional general-equilibrium feedbacks.

All code used to solve the models should be included in an appendix or submitted together with the report. The code should be sufficiently clear that the implementation of the main model equations and equilibrium conditions can be followed.

If you have used AI tools, include a short note describing how they were used, for example for programming assistance, debugging, or language editing. You remain responsible for understanding the model, verifying the numerical solution, and writing the economic interpretation.

10 Bonus Extensions

The additional extensions allow you to develop the model in different directions. Some add more realistic economic mechanisms, while others make the mathematical or numerical problem more challenging. You do not need to complete them in order, and you may choose extensions that fit your interests.

The additional parts are primarily intended as **bonus work for students aiming for higher grades**. They provide an opportunity to demonstrate a deeper understanding of the model, greater independence in implementation, and the ability to analyse more advanced economic mechanisms.

Each bonus extension is worth **1-3 points**. To receive the point, the extension must be implemented correctly, the numerical equilibrium or calculation must be verified, and the results must be presented and discussed.

For each extension, explain both the **mathematical solution** and the **economic interpretation**. Describe how the extension changes the model, which equations or equilibrium conditions are modified, and how you verify that the new solution is correct. You should also explain the economic mechanism behind the results: why do the variables change in the observed direction, and which prices, incentives, constraints, or general-equilibrium feedbacks generate the outcome?

You may receive at most **6 additional points**, even if you complete more than six extensions.

10.1 Extension A: A three-node network (1 point)

A third node is the smallest extension that can demonstrate genuine indirect network effects: a link can be improved between nodes 1 and 2 while node 3 is affected only through changed prices, demand, and production elsewhere in the network.

Let

$$\mathcal{N} = \{1, 2, 3\}.$$

The delivered-price equation remains

$$p_{ij} = p_i(1 + t_{ij}).$$

The household Armington price index becomes

$$P_j^H = \left[\sum_{i=1}^{3} \beta_{ij} p_{ij}^{1-\eta_H}\right]^{1/(1-\eta_H)}.$$

The household shares are

$$s_{ij}^H = \frac{\beta_{ij} p_{ij}^{1-\eta_H}}{\sum_{h=1}^{3} \beta_{hj} p_{hj}^{1-\eta_H}}.$$

If you include intermediate inputs, use the corresponding three-origin expressions for P_j^X and s_{ij}^X.

Normalize

$$w_1 = 1.$$

There are now two relative wages, w_2 and w_3.

A useful experiment is to reduce only

$$t_{12} \quad \text{and} \quad t_{21}$$

while leaving all links involving node 3 unchanged.

Investigate whether node 3 is affected even though no transport link connected to node 3 is directly improved.

10.2 Extension B: CES production as in the reference model (1 point)

The Cobb–Douglas production function in the compulsory part fixes cost shares. CES production adds a new behavioural margin: firms can change the mix of labour and intermediate goods when relative input prices change. The key question is whether this extra substitution dampens or amplifies transport-cost shocks.

Replace the Cobb–Douglas production function in Part V by a CES technology with labour and origin-specific intermediate goods:

$$Y_j = \phi_j \left[\delta_{lj} (L_j^d)^{-\rho_j} + \sum_i \delta_{ij} (X_{ij}^d)^{-\rho_j} \right]^{-1/\rho_j}. \tag{89}$$

The parameters satisfy

$$\delta_{lj} + \sum_i \delta_{ij} = 1. \tag{90}$$

The elasticity of substitution is

$$\sigma_j = \frac{1}{1+\rho_j}. \tag{91}$$

Derive or implement the corresponding unit-cost and conditional factor-demand functions.

Compare the results with the Cobb–Douglas production model. In particular, investigate how the elasticity of substitution σ_j between labour and intermediate goods changes the response to transport costs.

10.3 Extension C: Equivalent and compensating variation (1 point)

Utility or real-consumption changes are useful for ranking outcomes but are not measured in currency. Equivalent and compensating variation convert a welfare change into a monetary amount, which makes them especially useful for policy appraisal and comparisons with investment costs.

The compulsory part reports welfare using real consumption Q_j. In this extension, we instead construct monetary welfare measures. You can read more about these measured in the course book.

For a one-level CES Armington household, derive the expenditure function corresponding to

$$Q_j = \left[\sum_i \beta_{ij}^{1/\eta_H} Z_{ij}^{(\eta_H - 1)/\eta_H} \right]^{\eta_H/(\eta_H - 1)}.$$

Since the unit-expenditure function is P_j^H, total expenditure needed to obtain utility Q_j is

$$e_j(p, Q_j) = P_j^H Q_j. \tag{92}$$

For an initial equilibrium 0 and new equilibrium 1, define equivalent variation:

$$EV_j = e_j(p^0, Q_j^1) - e_j(p^0, Q_j^0). \tag{93}$$

Define compensating variation:

$$CV_j = e_j(p^1, Q_j^1) - e_j(p^1, Q_j^0). \tag{94}$$

Check and clearly state the sign convention used for EV_j and CV_j. Apply the same reduction in transport costs as in Part VI. For each region, calculate EV_j and CV_j, and compare these monetary welfare measures with the percentage change in real consumption Q_j. Discuss whether the three measures lead to the same qualitative conclusion about the welfare effect of the transport-cost reduction, and explain any differences.

11 Advanced Extensions

From Extension D onward, the extensions are intentionally less fully specified. You are expected to make and justify the additional modelling assumptions needed to formulate and solve the extended model. Each advanced extension is worth 2-3 points. To receive full credit, you must:

- clearly state the additional modelling assumptions you introduce;
- motivate why these assumptions are reasonable and appropriate for the question being studied;
- show how the assumptions modify the mathematical model and equilibrium conditions;
- implement and verify the resulting numerical solution;
- present and interpret the main results in economic terms;
- discuss how alternative modelling assumptions could affect the results and conclusions.

The purpose of the advanced extensions is not only to obtain a numerical solution, but to demonstrate that you can independently formulate a coherent economic model. In your report, distinguish clearly between assumptions inherited from the baseline model and assumptions introduced by you. Where several plausible modelling choices exist, explain why you selected your specification and discuss the consequences of that choice for both the mathematical solution and the economic results.

11.1 Advanced Extension D: Infrastructure Investments (2 points)

So far, transport costs have been treated as exogenous. In reality, reducing transport costs requires real resources for construction, and these resources must be financed. This extension makes infrastructure investment part of the general-equilibrium model.

The amount of resources devoted to construction determines the improvement in transport costs, while the location of construction and the distribution of financing determine how the costs and benefits are distributed across regions.

In general equilibrium, however, neither the money used to finance the project nor the real resources used for construction come from outside the economy. They must be taken from other uses. If more labour and goods are used for infrastructure construction, fewer resources are available for private consumption or other production. Likewise, higher taxes reduce households' disposable income and therefore their demand for other goods. The project therefore creates both benefits from lower transport costs and opportunity costs because scarce resources are redirected toward construction.

Let $C \geq 0$ denote the total amount of real resources devoted to an infrastructure project that improves the links between nodes 1 and 2. Assume that transport costs decline with construction resources according to

$$t_{12}(C) = t_{12}^0 e^{-\psi C}, \qquad t_{21}(C) = t_{21}^0 e^{-\psi C}, \tag{95}$$

where $\psi > 0$ measures the effectiveness of infrastructure investment.

This improvement is increasing in construction resources but exhibits diminishing returns. The first units of construction resources have a relatively large effect on transport costs, while further improvements become progressively more costly to achieve.

Construction can take place in either node. Let

$$C_k = \omega_k C, \qquad \omega_k \geq 0, \qquad \sum_k \omega_k = 1, \tag{96}$$

where ω_k is the share of construction activity undertaken in node k. Assume that construction in node k uses the locally produced good. Construction demand C_k must therefore be added to the goods-market equation for that node.

The monetary cost of the project is

$$I^G = \sum_k p_k C_k. \tag{97}$$

The project is financed by lump-sum taxes. Let

$$R_j = \gamma_j R, \qquad \gamma_j \geq 0, \qquad \sum_j \gamma_j = 1, \tag{98}$$

where γ_j determines the share of the total financing burden paid by households in node j. The government budget constraint is

$$R = \sum_k p_k C_k. \tag{99}$$

Disposable household income becomes

$$I_j = w_j \bar{L}_j - R_j. \tag{100}$$

Because producer prices are endogenous, the monetary construction cost and the required tax revenue must be determined jointly with wages, prices, production, and trade flows in the general equilibrium.

For any given value of C, solve the complete general equilibrium including the lower transport costs, construction demand, and tax financing. Measure welfare relative to the benchmark with $C = 0$, for example using equivalent variation[1],

$$EV_j(C). \tag{101}$$

Define aggregate welfare as

$$W(C) = \sum_j EV_j(C), \tag{102}$$

and determine the optimal amount of construction resources numerically:

$$C^* = \arg\max_{C \geq 0} W(C). \tag{103}$$

Thus, the size of the infrastructure project is determined by a trade-off. More construction resources reduce transport costs and generate economic benefits, but they also require additional production and higher taxes. Because equation (95) has diminishing returns, the marginal transport benefit declines as the project becomes larger.

Investigate how both the optimal project size and the regional welfare effects depend on who finances the project and where construction takes place. Compare, for example,

[1] If you have not completed Extension C, use Q_j as the welfare measure instead.

1. both nodes finance the investment and both nodes carry out construction;
2. both nodes finance the investment, but construction takes place only in node 1;
3. only node 1 finances the investment, while both nodes carry out construction;
4. only node 1 finances the investment and construction also takes place only in node 1.

For each scenario, calculate the optimal C^* and report the welfare effects in both nodes.
In your analysis, discuss:

1. Why is the optimal level of infrastructure investment finite? Explain the between the marginal benefit from lower transport costs and the marginal resource cost of construction.
2. How do the construction location and financing shares affect the distribution of welfare between the two nodes?
3. Distinguish between the three main channels: lower transport costs, income generated by construction activity, and the financing burden.
4. How does the investment-efficiency parameter ψ affect the optimal project size and welfare? What happens when infrastructure improvements become very expensive at the margin?

Clearly state and motivate any additional assumptions needed to close the model. Discuss how these assumptions affect both the mathematical solution and the economic conclusions.

11.2 Advanced Extension E: Capital as a second primary factor (2 points)

With labour as the only primary factor, all changes in factor scarcity are reflected in wages. Introducing capital adds a second factor market and allows shocks to affect wages and returns to capital differently. This also makes the regional incidence of transport-cost changes richer, since the benefits from improved market access may accrue differently to workers and capital owners. Allowing capital to move between regions introduces an additional spatial adjustment mechanism alongside trade.

Introduce regional capital K_j. A convenient Cobb–Douglas production technology is

$$Y_j = \phi_j \left(\frac{L_j^d}{\delta_{lj}}\right)^{\delta_{lj}} \left(\frac{K_j^d}{\delta_{kj}}\right)^{\delta_{kj}} \left(\frac{X_j^d}{\delta_{xj}}\right)^{\delta_{xj}}, \tag{104}$$

where

$$\delta_{lj} + \delta_{kj} + \delta_{xj} = 1. \tag{105}$$

Let r_j denote the rental rate of capital in region j. The corresponding unit-cost function is

$$c_j = \frac{1}{\phi_j} w_j^{\delta_{lj}} r_j^{\delta_{kj}} (P_j^X)^{\delta_{xj}}. \tag{106}$$

Under perfect competition and constant returns to scale,

$$p_j = c_j. \tag{107}$$

If

$$V_j = p_j Y_j \tag{108}$$

denotes the value of output, Cobb–Douglas cost shares imply

$$w_j L_j^d = \delta_{lj} V_j, \tag{109}$$

$$r_j K_j^d = \delta_{kj} V_j, \tag{110}$$

and

$$P_j^X X_j^d = \delta_{xj} V_j. \tag{111}$$

Hence factor demands can be written as

$$L_j^d = \frac{\delta_{lj} V_j}{w_j}, \tag{112}$$

and

$$K_j^d = \frac{\delta_{kj} V_j}{r_j}. \tag{113}$$

First assume that capital is immobile between regions. Each region has a fixed capital endowment

$$\bar{K}_j, \tag{114}$$

and regional capital-market clearing requires

$$K_j^d = \bar{K}_j. \tag{115}$$

Households now receive both labour and capital income. You must specify who owns the regional capital stock. A simple assumption is that households in region j own the capital located in that region. Household income then becomes

$$I_j = w_j \bar{L}_j + r_j \bar{K}_j. \tag{116}$$

Modify the household-demand and goods-market equations accordingly and solve the new general equilibrium.

Next, consider a version with mobile capital. If capital can move freely between regions, the regional rental rates must be equal in equilibrium:

$$r_1 = r_2 = r. \tag{117}$$

In this case, the regional capital stocks become endogenous:

$$K_1^d + K_2^d = \bar{K}, \tag{118}$$

where

$$\bar{K} = \bar{K}_1 + \bar{K}_2. \tag{119}$$

You must also specify how capital income is distributed across households when capital is mobile. For example, households may own fixed shares of the aggregate capital stock independent of where the capital is employed. Clearly state and motivate the ownership assumption you use.

Compare the following cases:

1. the baseline model without capital;
2. capital as an immobile regional factor;
3. capital that can move freely between the two regions.

Apply the same transport-cost reduction used in Part VI and compare the resulting changes in wages, rental rates, production, trade, factor allocation, and household welfare.

In your analysis, address the following questions:

1. How does introducing capital change the equilibrium conditions relative to the baseline model?
2. How are the effects of a transport-cost reduction divided between wages and returns to capital?
3. How does capital mobility change the regional allocation of production and the incidence of the transport-cost shock?
4. How does your assumption about capital ownership affect the regional distribution of welfare gains?
5. Compare the short-run case with immobile capital and the long-run case with mobile capital. What are the main economic differences?

Clearly state and motivate the additional assumptions needed to close the model, especially those concerning capital endowments, capital mobility, capital ownership, and the distribution of capital income. Discuss how these assumptions affect both the mathematical solution and the economic interpretation of the results.

11.3 Advanced Extension F: External Scale Economies, Agglomeration, Migration, and Multiple Equilibria (3 points)

The compulsory model assumes constant returns to scale and fixed regional labour endowments. In this extension, workers are allowed to move between regions and regional productivity depends on regional population. Together, these assumptions create a possible agglomeration mechanism: a region that attracts more workers becomes larger and more productive, which may make it still more attractive to workers.

The scale economy is assumed to be *external* to the individual firm. Each firm continues to have a constant-returns production technology and takes regional productivity as given. Perfect competition and the zero-profit condition can therefore still be used.

The purpose of the extension is to investigate how external scale economies affect the location of economic activity, whether several long-run spatial equilibria can exist, and how the stability of these equilibria changes when the strength of agglomeration changes.

Step 1: Introduce mobile population and external scale economies

Let total population be fixed:

$$N_1 + N_2 = \bar{N}, \tag{120}$$

and assume that each worker supplies one unit of labour, so regional labour supply is

$$\bar{L}_j = N_j. \tag{121}$$

Let regional scale be measured by regional population,

$$S_j = N_j. \tag{122}$$

Normalize scale relative to the symmetric benchmark population,

$$\bar{S}_j = \frac{\bar{N}}{2}. \tag{123}$$

Regional productivity is then

$$\phi_j = \bar{\phi}_j \left(\frac{N_j}{\bar{N}/2} \right)^{\mu}, \qquad \mu \geq 0, \tag{124}$$

where $\bar{\phi}_j$ is benchmark productivity and μ determines the strength of the external scale economy. With this normalization,

$$N_1 = N_2 = \frac{\bar{N}}{2}$$

implies

$$\phi_j = \bar{\phi}_j.$$

When $\mu = 0$, productivity is independent of regional population and the external scale effect disappears. When $\mu > 0$, a larger regional population raises productivity. This may represent

knowledge spillovers, thicker labour markets, better matching, shared suppliers, or other advantages of economic concentration.

For a given population allocation, the individual firm takes ϕ_j as given. The firm's production technology therefore remains constant returns to scale from the firm's perspective, and the competitive zero-profit condition remains

$$p_j = c_j. \tag{125}$$

At the regional level, however, equation (124) creates a possible positive feedback:

$$N_j \uparrow \Rightarrow \phi_j \uparrow \Rightarrow c_j \downarrow \Rightarrow p_j \downarrow \Rightarrow \text{regional real income may rise} \Rightarrow N_j \uparrow .$$

Step 2: Solve the general equilibrium for a given population allocation

Choose a candidate value of N_1. Population in region 2 is then determined by

$$N_2 = \bar{N} - N_1. \tag{126}$$

For this population allocation:

1. calculate regional labour supplies

$$\bar{L}_1 = N_1, \qquad \bar{L}_2 = N_2;$$

2. calculate regional productivity from equation (124);
3. solve the complete general equilibrium of the economy for prices, wages, production, trade flows, intermediate demand, and household consumption;
4. verify that the equilibrium residuals are sufficiently close to zero.

This gives a *conditional general equilibrium*: the economic equilibrium is solved taking the population allocation as temporarily given.

Repeat this calculation for many possible values of N_1.

Step 3: Measure regional attractiveness

A simple measure of real income per worker is

$$v_j = \frac{w_j}{P_j^H}, \tag{127}$$

where P_j^H is the household consumer price index.

If workers can move freely between regions, an interior long-run spatial equilibrium requires workers to be indifferent between the two locations:

$$v_1 = v_2. \tag{128}$$

With strong agglomeration forces, however, the model may generate very strong regional concentration. You may therefore introduce a dispersion force, such as congestion, housing costs, land scarcity, or other crowding effects.

One possible specification is

$$\widetilde{v}_j = \frac{w_j}{P_j^H \left(N_j/(\bar{N}/2)\right)^{\chi}}, \qquad \chi \geq 0, \tag{129}$$

where χ determines the strength of the dispersion force.

When $\chi = 0$,

$$\widetilde{v}_j = v_j.$$

When $\chi > 0$, a larger population makes a region less attractive through congestion or other crowding costs.

If equation (129) is used, the spatial equilibrium condition becomes

$$\widetilde{v}_1 = \widetilde{v}_2. \tag{130}$$

Step 4: Construct the migration function

For every candidate value of N_1, solve the conditional general equilibrium and define

$$G(N_1) = \ln \widetilde{v}_1(N_1) - \ln \widetilde{v}_2(\bar{N} - N_1), \tag{131}$$

or use v_j instead of $\widetilde{v}_j$ if no dispersion force is included.

The sign of $G(N_1)$ has a direct economic interpretation.

If

$$G(N_1) > 0,$$

real income is higher in region 1, so workers have an incentive to move toward region 1.

If

$$G(N_1) < 0,$$

real income is higher in region 2, so workers have an incentive to move toward region 2.

An interior spatial equilibrium satisfies

$$G(N_1^*) = 0. \tag{132}$$

Evaluate $G(N_1)$ over a sufficiently fine grid

$$N_1 \in [\varepsilon, \bar{N} - \varepsilon], \qquad \varepsilon > 0, \tag{133}$$

so that both regions remain populated and all equations remain well defined.

Plot $G(N_1)$ against N_1. Each crossing of the horizontal axis represents an interior spatial equilibrium.

Do not use a value of $G(N_1)$ unless the underlying general equilibrium has been successfully solved and its residuals are sufficiently close to zero.

Step 5: Determine the stability of each spatial equilibrium

To study local stability, consider the simple migration dynamic

$$\dot{N}_1 = \lambda G(N_1), \qquad \lambda > 0. \tag{134}$$

Workers therefore move toward the region offering higher real income.

At an equilibrium N_1^*,

$$G(N_1^*) = 0. \tag{135}$$

The equilibrium is locally stable if

$$G'(N_1^*) < 0, \tag{136}$$

because a small movement away from the equilibrium creates incentives that move population back toward it.

The equilibrium is locally unstable if

$$G'(N_1^*) > 0, \tag{137}$$

because a small deviation creates incentives for population to move still farther away.

Estimate $G'(N_1^*)$ numerically, for example using a finite difference, or infer stability from the direction in which $G(N_1)$ crosses the horizontal axis.

Step 6: Construct a bifurcation diagram

The main result of this extension should be summarized using a *bifurcation diagram.*

A bifurcation diagram shows how the equilibrium values of an endogenous variable change when a model parameter is varied. It is particularly useful when the number or stability of equilibria may change.

In this model, use

$$\mu$$

on the horizontal axis and the equilibrium population in region 1,

$$N_1^*,$$

on the vertical axis.

For each value of μ:

1. solve the model over a sufficiently fine grid of candidate values of N_1;
2. construct $G(N_1)$;
3. find *all* roots satisfying
$$G(N_1^*) = 0;$$
4. determine whether each equilibrium is stable or unstable from the sign of
$$G'(N_1^*);$$

5. record every equilibrium value N_1^*;
6. repeat the procedure for the next value of μ.

Plot all resulting equilibrium values N_1^* against μ. Clearly distinguish stable and unstable equilibria, for example by using solid lines or filled markers for stable equilibria and dashed lines or different markers for unstable equilibria.

The bifurcation diagram therefore summarizes three pieces of information simultaneously:

- the location of the spatial equilibria;
- the number of equilibria;
- the stability of each equilibrium.

In a symmetric two-region model,

$$N_1^* = N_2^* = \frac{\bar{N}}{2} \tag{138}$$

is a natural candidate equilibrium.

For sufficiently weak agglomeration forces, this symmetric equilibrium may be stable. As μ increases, it may lose stability and two asymmetric stable equilibria may emerge: one in which economic activity is concentrated mainly in region 1 and another in which activity is concentrated mainly in region 2.

A change in the number or stability of equilibria as a parameter varies is called a *bifurcation*.

A useful diagnostic is the slope of the migration function at the symmetric equilibrium,

$$G'\left(\frac{\bar{N}}{2}\right). \tag{139}$$

A critical value μ_c may exist such that

$$G'\left(\frac{\bar{N}}{2}; \mu_c\right) = 0. \tag{140}$$

At this value, the local stability of the symmetric equilibrium may change.

Figure 1 illustrates schematically the type of bifurcation diagram that can arise. The figure is only an illustration. Your bifurcation diagram must be generated from the numerical equilibria of your own model.

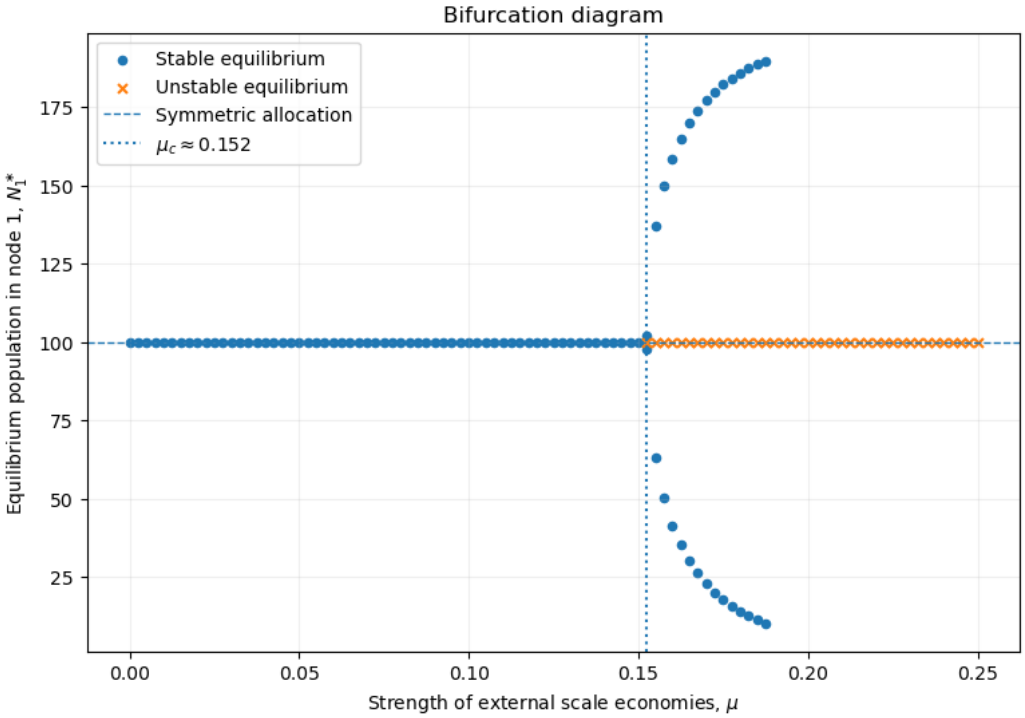


Figure 1: Schematic bifurcation diagram. The horizontal axis shows the strength of the external scale economy, μ, and the vertical axis shows the equilibrium population in region 1, N_1^*. Solid branches represent locally stable equilibria and the dashed branch represents an unstable equilibrium. At the critical value μ_c, the symmetric equilibrium changes stability and asymmetric spatial equilibria may emerge. The precise shape of the diagram is model dependent.

Step 7: Comparative experiments

Investigate a sufficiently fine range of values of μ, including

$$\mu = 0.$$

Construct the bifurcation diagram and identify, if possible, a critical value μ_c at which the symmetric equilibrium changes stability.

Then repeat the analysis under alternative parameter values.

At minimum, investigate:

- different values of the agglomeration parameter μ;
- different bilateral transport costs $t_{12} = t_{21}$;
- different values of the dispersion parameter χ, if the dispersion mechanism is included.

Compare the resulting bifurcation diagrams. In particular, investigate whether lower transport costs or stronger congestion shift the critical value μ_c, change the number of equilibria, or alter the degree of regional concentration.

Numerical checks

This extension may generate several equilibria and may become numerically difficult when the agglomeration force is strong.

For every conditional population allocation:

- verify the general-equilibrium residuals;
- use sensible starting values;
- try alternative starting values when necessary;
- do not interpret solver failure automatically as evidence that no equilibrium exists.

A solver may fail because the model is poorly conditioned, because several equilibria exist, because the initial values are poor, or because the chosen parameters generate very strong positive feedback.

Similarly, finding one root of $G(N_1)$ does not establish that it is the only spatial equilibrium. Search over the complete admissible population interval before drawing conclusions about uniqueness.

Analysis and discussion

In your report, present the migration functions and the bifurcation diagram and address the following questions:

1. How does a larger μ affect regional productivity, prices, wages, real income, and population allocation? Explain the agglomeration feedback mechanism.
2. Construct and interpret the bifurcation diagram. How many spatial equilibria exist for different values of μ, and which equilibria are stable or unstable? Can you identify a critical value μ_c at which the symmetric equilibrium changes stability?
3. Do external scale economies necessarily imply multiple equilibria? Explain why the answer depends on the strength of the agglomeration and dispersion forces.
4. How do lower transport costs affect the tendency toward regional concentration? Compare the bifurcation diagrams before and after a reduction in transport costs.
5. If a dispersion force is included, how does a larger χ affect the equilibrium structure? Can sufficiently strong congestion stabilize the symmetric equilibrium?
6. What happens when the agglomeration force becomes very strong? Does the model become sensitive to initial values or small parameter changes? Does the numerical solution become difficult to obtain?
7. If several stable equilibria exist, discuss *path dependence*. Can two economies with the same structural parameters but slightly different initial population distributions converge to different long-run spatial outcomes?

Clearly state all assumptions used in the extension, including the migration condition and any dispersion force. Distinguish carefully between economic multiplicity and numerical solver failure, and explain how the modelling assumptions affect both the mathematical solution and the economic interpretation of the results.

11.4 Advanced Extension G: Internal Scale Economies and Monopolistic Competition (3 points)

The compulsory model assumes perfect competition and constant returns to scale. Under these assumptions, firms charge a price equal to unit cost and earn zero economic profit. This structure is no longer appropriate when production involves internal economies of scale.

A common source of internal scale economies is a fixed production cost. A firm must use some resources before producing any output, but once the fixed cost has been paid, additional output can be produced at constant marginal cost. Average cost therefore falls as firm output increases.

In this extension, replace the representative competitive firm in each node by a number of monopolistically competitive firms. Each firm produces a differentiated variety. Consumers and firms value variety, firms charge a markup over marginal cost, and free entry ensures zero economic profit in the long-run equilibrium.

An important feature of the particular closure used below is that regional labour supplies are fixed and labour is the resource used for both fixed and variable production. As you will derive, these assumptions imply that the equilibrium number of firms in a region is determined by the regional labour endowment and structural parameters. Transport costs can affect firm size, prices, trade, and welfare, but they do not affect the number of firms in this version of the model. Understanding why this occurs is an important part of the extension.

Step 1: Variable production with a Cobb–Douglas technology

Let f index firms and j regions. For the *variable* part of production, assume that firm f in region j uses local labour and an Armington intermediate composite:

$$q_{fj} = \phi_j \left(\frac{L^v_{fj}}{\delta_{lj}} \right)^{\delta_{lj}} \left(\frac{X^v_{fj}}{\delta_{xj}} \right)^{\delta_{xj}}, \tag{141}$$

where L^v_{fj} is variable labour input and X^v_{fj} is the variable intermediate-input composite.

The cost shares satisfy

$$\delta_{lj} + \delta_{xj} = 1, \qquad \delta_{lj} > 0, \qquad \delta_{xj} > 0. \tag{142}$$

Let P^X_j denote the price index of the intermediate-input composite used by firms in region j. The variable unit-cost function is

$$c^v_j = \phi_j^{-1} w_j^{\delta_{lj}} (P^X_j)^{\delta_{xj}}. \tag{143}$$

The variable-production technology therefore has constant returns to scale. Producing twice as much output requires twice as much variable cost. Internal scale economies will instead arise from a fixed cost.

Step 2: Add the internal scale economy

Assume that every active firm must use a fixed amount of labour

$$F_j > 0. \tag{144}$$

Total labour used by firm f is therefore

$$L_{fj} = F_j + L^v_{fj}. \tag{145}$$

The total cost of producing q_{fj} units is

$$C_{fj}(q_{fj}) = w_j F_j + c^v_j q_{fj}. \tag{146}$$

Hence marginal cost is

$$MC_j = c^v_j, \tag{147}$$

while average cost is

$$AC_j(q_{fj}) = c^v_j + \frac{w_j F_j}{q_{fj}}. \tag{148}$$

Since

$$\frac{\partial AC_j}{\partial q_{fj}} < 0, \tag{149}$$

average cost falls as output per firm increases. The fixed requirement F_j therefore creates internal economies of scale.

Notice that the scale economy does not come from the Cobb–Douglas variable-production technology. The variable technology remains constant returns to scale. The internal scale economy arises because each active firm must incur the fixed labour cost F_j.

Step 3: Demand for differentiated firm varieties

Assume that the varieties produced by different firms are imperfect substitutes. Let

$$\varepsilon_j > 1 \tag{150}$$

denote the elasticity of substitution between firm varieties produced in region j.

With CES demand across firm varieties, each firm faces an isoelastic demand curve. Profit maximization therefore implies the constant-markup pricing rule

$$p^f_j = \frac{\varepsilon_j}{\varepsilon_j - 1} c^v_j, \tag{151}$$

where p^f_j is the factory-gate price charged by an individual firm in region j.

Define the markup factor

$$m_j = \frac{\varepsilon_j}{\varepsilon_j - 1}. \tag{152}$$

Then

$$p^f_j = m_j c^v_j. \tag{153}$$

A larger ε_j means that firm varieties are closer substitutes. Competition is therefore stronger and the markup is smaller.

Step 4: Free entry and equilibrium firm size

The profit of a representative firm is

$$\pi_j = \left(p_j^f - c_j^v\right) q_j - w_j F_j. \tag{154}$$

With free entry, positive profits attract entry and negative profits lead to exit. Long-run equilibrium therefore requires

$$\pi_j = 0. \tag{155}$$

Combining the markup rule and the zero-profit condition gives

$$q_j = \frac{(\varepsilon_j - 1) w_j F_j}{c_j^v}. \tag{156}$$

Thus free entry determines equilibrium output per firm conditional on wages and marginal cost.

The revenue of one firm is

$$R_j^f = p_j^f q_j = \varepsilon_j w_j F_j. \tag{157}$$

If total firm revenue in region j is V_j, the number of active firms is

$$n_j = \frac{V_j}{\varepsilon_j w_j F_j}. \tag{158}$$

At this stage, n_j appears to depend on regional revenue and wages. Below you will see that labour-market clearing places an additional restriction on the number of firms.

Step 5: The regional variety price index

Consumers and firms care not only about the total amount produced in a region but also about the number of available varieties.

If the n_j firms in region j are symmetric, the CES price index for varieties produced in that region is

$$\widetilde{p}_j = \left[n_j (p_j^f)^{1-\varepsilon_j}\right]^{1/(1-\varepsilon_j)}. \tag{159}$$

Equivalently,

$$\widetilde{p}_j = n_j^{-1/(\varepsilon_j - 1)} p_j^f. \tag{160}$$

This equation captures the love-of-variety effect. Holding individual firm prices constant, a larger number of firms lowers the price index of the regional composite.

Transport costs apply to the regional composites. The delivered price from origin i to destination j is

$$p_{ij} = (1 + t_{ij}) \widetilde{p}_i. \tag{161}$$

Use these delivered prices in the household and intermediate-input Armington equations from the compulsory model.

The model therefore contains two distinct substitution mechanisms:

- ε_i: substitution between individual firm varieties within origin i;
- η_H and η_X: Armington substitution between regional origin composites.

Step 6: Variable costs, intermediate demand, and labour demand

Because firms charge a markup, total revenue is no longer equal to variable production cost.
Define

$$\lambda_j = \frac{c_j^v}{p_j^f} = \frac{\varepsilon_j - 1}{\varepsilon_j}. \tag{162}$$

The fraction λ_j of revenue is used to cover variable production costs, while the remaining fraction finances the fixed labour cost.
If V_j denotes total firm revenue in region j, total variable production cost is

$$C_j^v = \lambda_j V_j. \tag{163}$$

Because the variable-production technology is Cobb–Douglas, the variable cost shares are fixed. Intermediate-input expenditure is therefore

$$E_j^X = \delta_{xj} \lambda_j V_j, \tag{164}$$

and variable labour expenditure is

$$E_j^{L,v} = \delta_{lj} \lambda_j V_j. \tag{165}$$

Variable labour demand is therefore

$$L_j^v = \frac{\delta_{lj} \lambda_j V_j}{w_j}. \tag{166}$$

Fixed labour demand is

$$n_j F_j. \tag{167}$$

Total labour demand becomes

$$L_j^d = \frac{\delta_{lj} \lambda_j V_j}{w_j} + n_j F_j. \tag{168}$$

Labour-market clearing requires

$$L_j^d = \bar{L}_j. \tag{169}$$

An important implication for the number of firms. The combination of free entry and labour-market clearing has a strong implication in this model.
Using

$$n_j = \frac{V_j}{\varepsilon_j w_j F_j}$$

in equation (168) gives

$$\bar{L}_j = \frac{V_j}{w_j}\left[\delta_{lj}\frac{\varepsilon_j - 1}{\varepsilon_j} + \frac{1}{\varepsilon_j}\right]. \tag{170}$$

Solving for V_j/w_j and substituting back into the free-entry condition gives

$$\boxed{n_j = \frac{\bar{L}_j}{F_j\left[1 + \delta_{lj}(\varepsilon_j - 1)\right]}.} \tag{171}$$

This is an important structural result of the model.

With fixed regional labour supply, the equilibrium number of firms depends on

$$\bar{L}_j, \qquad F_j, \qquad \delta_{lj}, \qquad \varepsilon_j,$$

but not directly on transport costs, productivity, Armington shares, or market access.

The economic reason can also be seen at the firm level. Variable labour expenditure per firm is

$$\delta_{lj}\lambda_j R_j^f = \delta_{lj}(\varepsilon_j - 1)w_j F_j. \tag{172}$$

Hence variable labour per firm is

$$L_j^{v,f} = \delta_{lj}(\varepsilon_j - 1)F_j. \tag{173}$$

Including the fixed labour requirement, each active firm uses

$$L_j^f = F_j\left[1 + \delta_{lj}(\varepsilon_j - 1)\right]. \tag{174}$$

Since regional labour supply is fixed and fully employed, the number of firms is simply regional labour supply divided by labour use per firm.

This does *not* imply that transport costs are irrelevant. A transport-cost change can still affect

$$w_j, \qquad V_j, \qquad c_j^v, \qquad p_j^f, \qquad q_j, \qquad \widetilde{p}_j, \qquad s_{ij}^H, \qquad s_{ij}^X, \qquad P_j^H, \qquad Q_j.$$

The important distinction is that transport integration changes the *scale of each firm* and the allocation of expenditure and production, but not the *number of firms* when $\bar{L}_j$, F_j, δ_{lj}, and ε_j are fixed.

Step 7: Modify the input–output block

Household income remains

$$I_j = w_j \bar{L}_j. \tag{175}$$

Free entry implies zero economic profit, so households do not receive profit income in the benchmark specification.

Household demand is calculated using the household Armington shares s_{ij}^H.

Intermediate-input expenditure by firms in destination j is

$$E_j^X = \delta_{xj}\lambda_j V_j. \tag{176}$$

Using intermediate Armington shares s_{ij}^X, expenditure on intermediate inputs from origin i is

$$E_{ij}^X = s_{ij}^X \delta_{xj} \lambda_j V_j. \tag{177}$$

The regional value-market equations become

$$V_i = \sum_j \left[s_{ij}^H I_j + s_{ij}^X \delta_{xj} \lambda_j V_j \right]. \tag{178}$$

Define

$$b_i = \sum_j s_{ij}^H I_j \tag{179}$$

and

$$M_{ij} = s_{ij}^X \delta_{xj} \lambda_j. \tag{180}$$

Then

$$V = b + MV. \tag{181}$$

Conditional on prices, wages, and the number of firms, solve

$$(I - M)V = b. \tag{182}$$

Thus the input–output block has the same structure as in Part V, but only the variable-cost share of firm revenue generates intermediate-input demand.

Step 8: Close and solve the equilibrium

For the two-region version, one possible nonlinear unknown vector is

$$x = \left(p_1^f, p_2^f, w_2, n_1, n_2 \right), \tag{183}$$

with

$$w_1 = 1 \tag{184}$$

as the numeraire.

For every trial vector x:

1. Calculate regional variety price indices:

$$\widetilde{p}_i = n_i^{-1/(\varepsilon_i - 1)} p_i^f.$$

2. Calculate delivered prices:

$$p_{ij} = (1 + t_{ij}) \widetilde{p}_i.$$

3. Calculate household and intermediate Armington price indices and sourcing shares.

4. Calculate household incomes:

$$I_j = w_j \bar{L}_j.$$

5. Construct b and M, and solve

$$(I - M)V = b.$$

6. Calculate the Cobb–Douglas variable unit costs:

$$c_j^v = \phi_j^{-1} w_j^{\delta_{lj}} (P_j^X)^{\delta_{xj}}.$$

7. Calculate total labour demand using equation (168).

8. Evaluate the markup conditions:

$$R_j^p = p_j^f - \frac{\varepsilon_j}{\varepsilon_j - 1} c_j^v. \tag{185}$$

9. Evaluate the free-entry conditions:

$$R_j^n = V_j - n_j \varepsilon_j w_j F_j. \tag{186}$$

10. Use one independent labour-market condition, for example

$$R^L = L_2^d - \bar{L}_2. \tag{187}$$

The five residuals

$$R(x) = \left(R_1^p, R_2^p, R_1^n, R_2^n, R^L\right) \tag{188}$$

determine the five nonlinear unknowns.

After solving, verify the omitted labour-market condition,

$$L_1^d - \bar{L}_1 \approx 0, \tag{189}$$

as well as

- the markup conditions;
- the free-entry conditions;
- zero firm profits;
- equation (171);
- the input–output identities;
- goods-market clearing including iceberg resource use.

All prices, wages, outputs, and numbers of firms must remain positive. Solving for logarithms of these variables can therefore improve numerical robustness.

Step 9: Numerical experiments

Use the following benchmark values unless otherwise stated:

$$F_1 = F_2 = 2, \qquad \varepsilon_1 = \varepsilon_2 = 5. \tag{190}$$

Together with the benchmark parameters from Part V, first solve a symmetric equilibrium with

$$\bar{L}_1 = \bar{L}_2 = 100, \qquad t_{12} = t_{21} = 0.20. \tag{191}$$

Verify that the two regions have identical equilibrium outcomes and that the markup, free-entry, labour-market, input–output, and goods-market residuals are sufficiently close to zero.

Also verify equation (171) directly.

Then perform the following experiments.

Experiment 1: Fixed costs. Vary F_j while holding other parameters fixed.

Study how a larger fixed cost affects

- the number of firms;
- output per firm;
- firm prices;
- the regional variety price index;
- household welfare.

Use equation (171) to explain why a larger fixed labour requirement generates fewer firms.

Explain also why the reduction in the number of firms creates a loss of variety.

Experiment 2: Elasticity between firm varieties. Vary ε_j.

Study the effects on

- the markup;
- firm size;
- the number of firms;
- the regional variety price index.

Explain why a larger ε_j lowers the markup and use equation (171) to explain its effect on the number of firms.

When varying ε_j, interpret comparisons of markups, firm size, and firm numbers directly. Be more cautious when comparing welfare levels across different values of ε_j, since changing ε_j also changes the CES aggregator over firm varieties and therefore the normalization of the regional variety price index.

Experiment 3: Transport-cost reduction. Apply the same symmetric transport-cost reduction used in Part VI:

$$t_{12} = t_{21} : 0.20 \to 0.10. \tag{192}$$

Compare the equilibrium before and after the reduction and report changes in

- the number of firms;
- output per firm;
- firm prices;
- regional variety price indices;
- bilateral trade;
- regional wages;
- regional production values;
- household price indices;
- real household consumption.

Before solving the new equilibrium, use equation (171) to predict what should happen to n_1 and n_2.

Verify numerically that the transport-cost reduction leaves the number of firms unchanged.

Then explain how lower transport costs can nevertheless change firm output, marginal costs, firm prices, sourcing patterns, trade, consumer prices, and welfare.

In particular, distinguish between

$$\text{the extensive margin: number of firms}$$

and

$$\text{the intensive margin: output per firm.}$$

In this model, transport integration operates through the intensive margin rather than through transport-induced entry and exit.

Experiment 4: Market-size asymmetry. Introduce asymmetric labour endowments. Use, for example,

$$\bar{L}_1 = 120, \qquad \bar{L}_2 = 80. \tag{193}$$

Keep

$$t_{12} = t_{21} = 0.20.$$

Investigate how the difference in regional size affects

- regional wages;
- regional production values;
- the number of firms;
- output per firm;
- regional variety price indices;
- household price indices;
- real household consumption.

Use equation (171) to predict the number of firms in each region before solving the complete equilibrium.

If F_j, δ_{lj}, and ε_j are identical across regions, show that

$$\frac{n_1}{n_2} = \frac{\bar{L}_1}{\bar{L}_2}. \tag{194}$$

Thus, in this model, a larger regional labour endowment supports more firms and therefore more varieties.

Because the regions now have different populations, also compare real income per worker,

$$v_j = \frac{w_j}{P_j^H}. \tag{195}$$

Distinguish the effect of market size on the number of varieties from the effects of equilibrium wages, prices, and trade.

Experiment 5: Test the invariance result. The independence of firm numbers from transport costs is not a consequence of symmetry. It follows from the combination of fixed regional labour supply, Cobb–Douglas variable production, fixed labour costs, and free entry.

Continue to use the asymmetric economy

$$\bar{L}_1 = 120, \qquad \bar{L}_2 = 80,$$

and reduce symmetric transport costs from

$$t_{12} = t_{21} = 0.20$$

to

$$t_{12} = t_{21} = 0.10.$$

For both equilibria, report

- n_1 and n_2;
- output per firm q_1 and q_2;
- firm prices p_1^f and p_2^f;
- regional variety price indices $\widetilde{p}_1$ and $\widetilde{p}_2$;
- regional wages w_1 and w_2;
- regional production values V_1 and V_2;
- household price indices P_1^H and P_2^H;
- real household consumption Q_1 and Q_2;
- real income per worker v_1 and v_2.

Verify that n_1 and n_2 are unchanged even though other equilibrium variables respond to the transport improvement.

Explain how V_j and w_j adjust consistently with

$$n_j = \frac{V_j}{\varepsilon_j w_j F_j}.$$

This experiment illustrates an important general lesson in economic modelling: whether a variable responds to a policy shock depends not only on the economic intuition but also on the closure assumptions of the model.

Step 10: Compare with the competitive model

Compare the monopolistically competitive equilibrium with the competitive model from Part V.

The monopolistically competitive model introduces several mechanisms absent under perfect competition:

- fixed production costs;
- markups over marginal cost;
- endogenous firm scale;
- an endogenous number of varieties;
- a love-of-variety effect on regional price indices.

It is useful to distinguish two different mechanisms in this particular model.

First, regional size and structural parameters determine the number of firms:

$$\left(\bar{L}_j, F_j, \delta_{lj}, \varepsilon_j\right) \to n_j \to \text{number of varieties.}$$

Second, transport costs operate through delivered prices and general-equilibrium adjustments:

$$t_{ij} \to p_{ij} \to \text{sourcing and trade} \to P_j^X \to c_j^v \to p_j^f,\ q_j \to P_j^H,\ Q_j.$$

The transport channel therefore affects firm scale but not firm numbers under the present closure.

When comparing the monopolistically competitive model with the competitive model, focus primarily on the response to the same transport-cost shock.

Do not mechanically compare the levels of Q_j across the two market structures. The monopolistically competitive model contains an additional CES aggregator over firm varieties, so the level of the regional variety price index depends on the normalization of the love-of-variety term.

Percentage changes within each model are therefore generally more informative.

Also note that monopolistic competition does not automatically imply a larger welfare response to a transport-cost reduction. Since transport costs do not change the number of firms in this specification, there is no additional transport-induced variety margin. Differences relative to the competitive model instead arise through markups, fixed-cost resource use, variable production costs, and the resulting general-equilibrium responses.

Analysis and discussion

In your report, address the following questions:

1. Why can the competitive pricing condition $p_j = c_j$ no longer be used when firms face fixed costs?

2. Why does free entry imply zero economic profit even though firms charge a markup above marginal cost?

3. Derive equilibrium output per firm from the markup and zero-profit conditions.

4. Derive equation (171). Why is the equilibrium number of firms determined by the regional labour endowment and structural parameters?

5. How does F_j affect firm size and the number of active firms? Explain the resulting love-of-variety effect.

6. How does ε_j affect the markup, equilibrium firm size, and the number of firms?

7. Why does a reduction in transport costs leave n_j unchanged in this model? Which equilibrium variables can still respond?

8. Explain the distinction between the extensive margin n_j and the intensive margin q_j. Which margin responds to transport integration here?

9. Does a larger regional labour endowment support more firms? Explain why.

10. Compare the welfare response to lower transport costs under perfect competition and monopolistic competition. Which additional mechanisms are present under monopolistic competition?

11. What modelling assumption would have to be changed for transport improvements to affect the regional number of firms? Discuss at least one possibility, such as labour mobility, elastic labour supply, an additional labour-using sector, or another primary factor.

12. Can the model become numerically or economically poorly behaved for very large fixed costs or very low substitution elasticities? Explain why.

Clearly state and motivate all additional assumptions needed to close the model, especially the specification of fixed costs, the treatment of differentiated varieties, the entry condition, and the treatment of firm ownership and profit income.

As throughout the laboratory, do not interpret a solver convergence message as sufficient evidence that an equilibrium has been found. Verify all equilibrium and accounting residuals independently before interpreting the results.

11.5 Advanced Extension H: A Shortest-Path Transport Network (3 points)

The compulsory spatial model assumes that bilateral transport costs t_{ij} are given directly for every origin–destination pair. Real transport systems are instead organized as networks. Goods may have to pass through several intermediate regions before reaching their destination, and an improvement to one physical link can therefore affect transport costs between many origin–destination pairs simultaneously.

Spatial computable general equilibrium models have been used to study the wider economic effects of transport infrastructure investments in Sweden.[2]

In this extension, replace the exogenous bilateral transport-cost matrix by a transport network based on the eight Swedish NUTS 2 regions. The purpose is to study how the position of a transport link in the network affects both the *total economic benefit* of improving that link and the *distribution of benefits* across regions.

The network and shortest-path formulation used below are stylized and are not intended as a replication of Sundberg2009. The coordinates and link distances are chosen to reproduce the approximate geography of Sweden and should not be interpreted as an exact representation of the Swedish road or railway system.

Step 1: Define the regions

Let the set of regions be

$$\mathcal{N} = \{\text{SE11}, \text{SE12}, \text{SE21}, \text{SE22}, \text{SE23}, \text{SE31}, \text{SE32}, \text{SE33}\}.$$

Use the approximate coordinates in Table 1.

Table 1: Approximate coordinates for the Swedish NUTS 2 regions.

Code	Region	Latitude	Longitude
SE33	Övre Norrland	65.20	20.00
SE32	Mellersta Norrland	62.80	16.80
SE31	Norra Mellansverige	60.70	14.60
SE12	Östra Mellansverige	59.20	16.00
SE11	Stockholm	59.33	18.07
SE23	Västsverige	57.90	12.50
SE21	Småland med öarna	57.30	15.20
SE22	Sydsverige	56.10	13.40

These coordinates are used both to visualize the network and to calculate approximate relative distances between connected regions.

Step 2: Construct the physical transport network

Represent the transport system by a graph

$$\mathcal{G} = (\mathcal{N}, E), \tag{196}$$

where the nodes are NUTS 2 regions and the edges are direct transport links.

[2] See Marcus Sundberg, *Economic Effects of the Öresund Bridge: A Spatial Computable General Equilibrium Analysis*, KTH Royal Institute of Technology, 2009. `https://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-10540`.

Use the network shown in Figure 2.

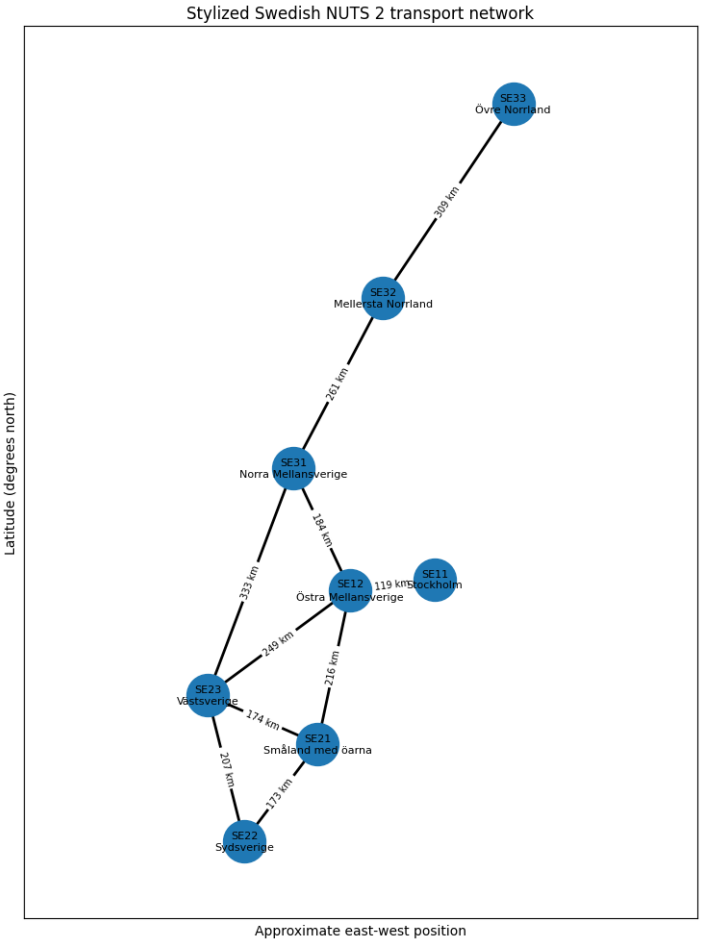


Figure 2: Stylized transport network based on the Swedish NUTS 2 regions. Nodes represent regions and edges represent direct transport links. Node positions and link lengths are approximate and are intended to reproduce the broad geographical structure of Sweden.

The benchmark physical links and approximate distances are given in Table 2.

Table 2: Benchmark physical transport links and approximate distances.

Link	Approximate distance (km)
SE33–SE32	309.0
SE32–SE31	260.6
SE31–SE12	184.1
SE31–SE23	333.3
SE12–SE11	118.5
SE12–SE23	249.2
SE12–SE21	216.4
SE23–SE21	174.1
SE23–SE22	207.4
SE21–SE22	172.8

The graph is undirected. Thus every physical link can be used in both directions.

The network is deliberately incomplete. Some origin–destination pairs therefore require goods to pass through one or more intermediate regions.

Step 3: Convert distance into iceberg edge costs

Each physical edge $e \in E$ has an iceberg transport markup

$$\tau_e \geq 0. \tag{197}$$

Let d_e denote the approximate distance of edge e, measured in kilometres. Generate the benchmark edge cost from

$$1 + \tau_e = \exp(\kappa d_e), \tag{198}$$

or equivalently

$$\tau_e = \exp(\kappa d_e) - 1. \tag{199}$$

Use

$$\kappa = 0.00030. \tag{200}$$

Thus a longer physical link receives a larger iceberg transport cost.

The physical edge cost τ_e should not be confused with the bilateral origin–destination transport cost t_{ij}. The latter depends on the complete least-cost route through the network.

Step 4: Calculate shortest paths and the bilateral transport-cost matrix

If a route from origin i to destination j consists of the set of edges

$$\mathcal{P} : i \to j,$$

its total iceberg factor is

$$\prod_{e \in \mathcal{P}} (1 + \tau_e). \tag{201}$$

The effective bilateral transport cost is therefore

$$1 + t_{ij} = \min_{\mathcal{P}: i \to j} \prod_{e \in \mathcal{P}} (1 + \tau_e), \qquad i \neq j, \tag{202}$$

with

$$t_{ii} = 0. \tag{203}$$

Most shortest-path algorithms use additive rather than multiplicative edge costs. Use

$$c_e = \ln(1 + \tau_e). \tag{204}$$

Then

$$\ln(1 + t_{ij}) = \min_{\mathcal{P}: i \to j} \sum_{e \in \mathcal{P}} c_e. \tag{205}$$

You may therefore use a standard shortest-path algorithm such as Dijkstra's algorithm.
After obtaining the shortest-path distance D_{ij}, recover the bilateral iceberg cost from

$$1 + t_{ij} = \exp(D_{ij}). \tag{206}$$

Calculate the shortest path for every ordered origin–destination pair and construct the complete matrix

$$T = [t_{ij}]. \tag{207}$$

Report the matrix and give several examples of shortest routes involving intermediate nodes.

Step 5: Economic benchmark

Use the benchmark economic parameters in Table 3.

Table 3: Benchmark economic parameters for the network experiment.

Parameter	Value	Interpretation
$\bar{L}_j$	100	Labour endowment in each region
ϕ_j	1	Productivity in each region
δ_{lj}	0.60	Labour cost share
δ_{xj}	0.40	Intermediate-input cost share
η_H	4	Household Armington elasticity
η_X	4	Intermediate-input Armington elasticity
β_{jj}	0.55	Household home weight
γ_{jj}	0.55	Intermediate-input home weight
β_{ij}, $i \neq j$	0.45/7	Foreign household weight
γ_{ij}, $i \neq j$	0.45/7	Foreign intermediate-input weight
κ	0.00030	Distance-to-edge-cost parameter
θ	0.50	Benchmark link-cost reduction

The Armington weights satisfy

$$\sum_i \beta_{ij} = 1, \qquad \sum_i \gamma_{ij} = 1.$$

Use one regional wage as numeraire. For example, if SE33 is region 1, set

$$w_{\mathrm{SE33}} = 1. \tag{208}$$

The symmetric factor endowments and technologies are intentional. They make the network structure the main source of spatial differences in the benchmark.

Step 6: Insert the network into the SCGE model

Use the shortest-path transport costs in the delivered-price equation,

$$p_{ij} = p_i(1 + t_{ij}). \tag{209}$$

For households, calculate the Armington price index

$$P_j^H = \left[\sum_i \beta_{ij} p_{ij}^{1-\eta_H} \right]^{1/(1-\eta_H)} \tag{210}$$

and expenditure shares

$$s_{ij}^H = \frac{\beta_{ij} p_{ij}^{1-\eta_H}}{\sum_h \beta_{hj} p_{hj}^{1-\eta_H}}. \tag{211}$$

For intermediate inputs, use

$$P_j^X = \left[\sum_i \gamma_{ij} p_{ij}^{1-\eta_X} \right]^{1/(1-\eta_X)} \tag{212}$$

and

$$s_{ij}^X = \frac{\gamma_{ij} p_{ij}^{1-\eta_X}}{\sum_h \gamma_{hj} p_{hj}^{1-\eta_X}}. \tag{213}$$

Production in each region is

$$Y_j = \phi_j \left(\frac{L_j^d}{\delta_{lj}} \right)^{\delta_{lj}} \left(\frac{X_j^d}{\delta_{xj}} \right)^{\delta_{xj}}, \tag{214}$$

with

$$\delta_{lj} + \delta_{xj} = 1. \tag{215}$$

The competitive unit-cost condition is

$$p_j = \phi_j^{-1} w_j^{\delta_{lj}} (P_j^X)^{\delta_{xj}}. \tag{216}$$

Household income is

$$I_j = w_j \bar{L}_j. \tag{217}$$

Let

$$V_j = p_j Y_j. \tag{218}$$

The input–output system is

$$V = b + MV, \tag{219}$$

where

$$b_i = \sum_j s_{ij}^H I_j \tag{220}$$

and

$$M_{ij} = s_{ij}^{X} \delta_{xj}. \tag{221}$$

Labour demand is

$$L_j^d = \frac{\delta_{lj} V_j}{w_j}. \tag{222}$$

Solve the complete general equilibrium for all regional wages, producer prices, production values, trade flows, and household consumption.

After solving, verify:

- the nonlinear equilibrium residuals;
- all labour-market conditions;
- the physical goods-market equations;
- the input–output identities.

Step 7: Improve one physical transport link

Choose one existing physical edge

$$e^* \in E \tag{223}$$

and reduce its transport cost according to

$$\tau_{e^*}^1 = (1 - \theta)\tau_{e^*}^0. \tag{224}$$

For the benchmark experiment use

$$\theta = 0.50. \tag{225}$$

Thus the iceberg markup on the selected link is reduced by 50%.

Because links are bidirectional, the improvement applies in both directions.

Keep all other physical edge costs unchanged.

After changing the selected edge cost, recompute the shortest paths for *all* origin–destination pairs and construct the new matrix

$$T^1 = [t_{ij}^1]. \tag{226}$$

Do not assume that only the two endpoint regions are affected.

Step 8: Identify network effects

For each link improvement, determine:

- which origin–destination pairs obtain a lower t_{ij};
- which shortest routes remain unchanged;
- which shortest routes change;
- how many benchmark shortest paths use the improved edge;

- how large the reduction in effective bilateral transport costs is.

Distinguish between two cases.

An investment may lower the cost of an already optimal route without changing the shortest path. Alternatively, it may make another route sufficiently attractive that the shortest path changes.

A link can therefore be economically important even if it generates few route changes.

Step 9: Re-solve the general equilibrium

Using T^1, solve the full SCGE model again.

For every region j, report at least

- the change in the regional wage;
- the change in production value;
- the change in the household price index;
- the change in real household consumption;
- important changes in bilateral trade.

Use

$$Q_j = \frac{I_j}{P_j^H} \tag{227}$$

as the regional real-consumption measure.

Define the welfare benefit received by region j as

$$\Delta Q_j = Q_j^1 - Q_j^0. \tag{228}$$

Step 10: Compare all single-link improvements

Repeat the experiment for *every* physical link in Table 2, improving one edge at a time.

Use the same proportional improvement

$$\theta = 0.50$$

for every project.

For each edge e, calculate the total welfare benefit

$$B_e = \sum_j \Delta Q_j. \tag{229}$$

Rank all links according to B_e.

Also calculate, for each physical edge, the number of benchmark shortest paths that use the edge.

Compare the two rankings.

A link with high network centrality need not necessarily have the highest economic value. The welfare effect also depends on the size of the transport-cost reduction, the trade flows affected, substitution between origins, intermediate-input linkages, and general-equilibrium adjustments.

Step 11: Study the distribution of benefits

For every link improvement, report the full regional welfare vector

$$(\Delta Q_1, \ldots, \Delta Q_8)\,. \tag{230}$$

Calculate each region's share of the total benefit,

$$s_j^B = \frac{\Delta Q_j}{\sum_k \Delta Q_k}. \tag{231}$$

Present these results in a table or heat map with

- improved physical links on one axis;
- NUTS 2 regions on the other axis;
- regional welfare gains in the cells.

Use these results to distinguish between

total efficiency

and

regional distribution.

Two infrastructure projects may generate similar aggregate benefits while distributing those benefits very differently across regions.

Also investigate whether regions that are not endpoints of the improved edge can obtain substantial benefits or losses.

Step 12: Investigate the size of the improvement

For a small number of important links, repeat the experiment using several values of

$$\theta,$$

for example

$$\theta \in \{0.10, 0.25, 0.50, 0.75\}. \tag{232}$$

Plot the total benefit B_e against the size of the transport improvement.

Investigate whether the welfare effect is approximately proportional to the size of the cost reduction or whether the response becomes nonlinear.

In particular, identify whether larger improvements cause shortest routes to switch. Route switching may generate discrete changes in economic outcomes.

Analysis and discussion

In your report, include the network figure, the benchmark shortest-path transport-cost matrix, a ranking of alternative link improvements, and the regional distribution of benefits.

Address the following questions:

1. Explain the difference between the physical edge cost τ_e and the bilateral transport cost t_{ij}. How is t_{ij} generated from the network?
2. Which physical links are used by the largest number of benchmark shortest paths?
3. Which single-link improvements generate the largest total welfare benefits? Explain why these links are important in the network.
4. Are the links with the largest number of shortest paths also the links with the largest economic benefits? If not, explain why.
5. Which investments cause shortest routes to change? Are route changes necessary for an investment to generate a large welfare effect?
6. How are the benefits distributed across the Swedish NUTS 2 regions? Which projects generate concentrated benefits and which generate more widely distributed benefits?
7. Can regions that are not endpoints of an improved edge gain from the investment? Explain the network and general-equilibrium mechanisms.
8. Do any regions lose even when the total welfare effect is positive? If so, explain why.
9. Compare the direct effect on shortest-path transport costs with the final general-equilibrium effects on wages, production, trade, consumer prices, and welfare.
10. How does the ranking of projects change when the size of the link improvement is varied?

Clearly distinguish between the physical transport network $\mathcal{G} = (\mathcal{N}, E)$, the shortest-path transport-cost matrix $T = [t_{ij}]$, and the resulting general-equilibrium economic outcomes.

The shortest-path calculation must be repeated after every change to the physical network.

As throughout the laboratory, do not interpret a solver convergence message as sufficient evidence that the equilibrium has been solved. Verify that all relevant equilibrium residuals are sufficiently close to zero before interpreting the results.